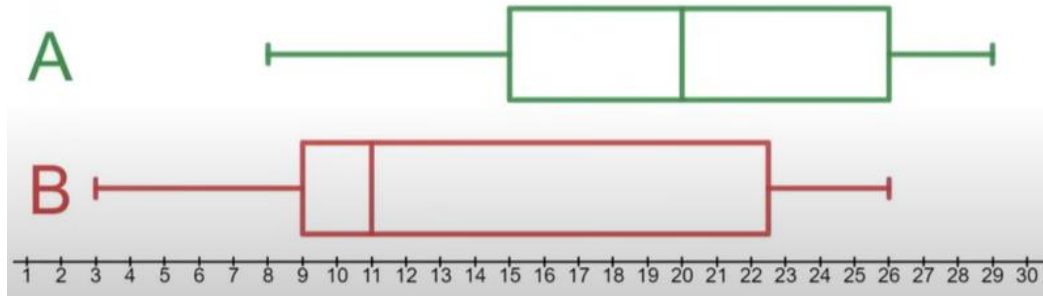


Hard questions 5

1.



Each of the box plots shown summarizes a data set. If the two data sets are combined into one data set, what is the range of the combined data set?

2.

$$3x^2 + 18x + 3y^2 - 6y - 15 = 0$$

The equation above gives the graph of a circle in the xy -plane. If the circle is inscribed in a square, what is the area of the square?

3.

Movie Theater Chain	Number of Screens (thousands)
AMC	11
Cinepolis	5
Wanda	x
Cinemark	6
Regal	7
CGV	3
Cineworld	9
Cinemex	2

A table of some movie theater chains is shown above. A movie theater chain is considered a big chain if it has more than 6,000 screens. The ratio of big chain screens to all screens is k to 5. If Wanda is a small chain, which expression represents k in terms of x ?

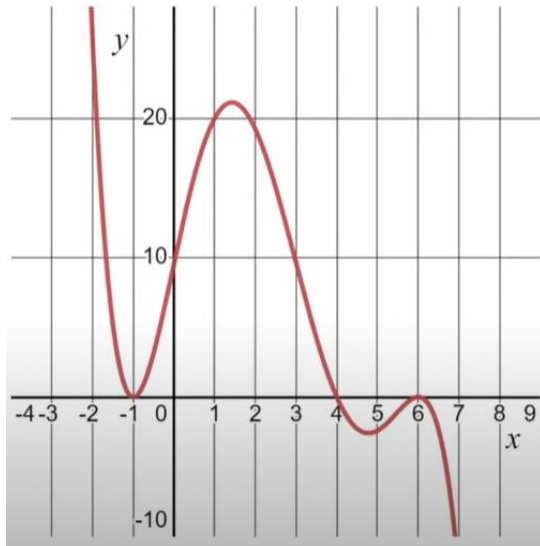
- A. $\frac{135}{43+x}$
- B. $\frac{43+x}{135}$
- C. $\frac{135+x}{43}$
- D. $\frac{43}{135+x}$

4.

$$\frac{3}{2x+7} - \frac{4x-6}{5}$$

The above expression is equivalent to $\frac{ax^2+bx+c}{10x+35}$ where a , b , and c are integers. What is the value of $a + b + c$?

5.



Which of the following could be the equation of the graph above?

(Try without using Desmos first!)

A) $y = -\frac{1}{15}(x - 6)(x - 4)^2(x + 1)$

B) $y = -\frac{1}{15}(x - 6)^2(x - 4)(x + 1)^2$

C) $y = -\frac{1}{15}(x + 6)(x + 4)^2(x - 1)$

D) $y = -\frac{1}{15}(x + 6)^2(x + 4)(x - 1)^2$

6.

In right triangle XYZ , the length of side XZ is 8, the measure of angle Z is 90° , and the measure of angle Y is 56° . Which of the following represents the length of side YZ ?

A) $8 \sin(56^\circ)$

B) $8 \tan(56^\circ)$

C) $\frac{8}{\sin(56^\circ)}$

D) $\frac{8}{\tan(56^\circ)}$